

CONCEPT CHECK 45.2

1. What are the roles of oxytocin and prolactin in regulating the mammary glands?
2. How do the two fused glands of the pituitary gland differ in function?
3. **WHAT IF?** Propose an explanation for why defects in a particular hormone cascade pathway observed in patients typically affect the final gland in the pathway rather than the hypothalamus or pituitary.
4. **WHAT IF?** Lab tests of two patients, each diagnosed with excessive thyroid hormone production, revealed an elevated level of TSH in one but not the other. Was the diagnosis of one patient necessarily incorrect? Explain.

For suggested answers, see Appendix A.

CONCEPT 45.3

Endocrine glands respond to diverse stimuli in regulating homeostasis, development, and behavior

In the remainder of this chapter, we'll focus on endocrine function in homeostasis, development, and behavior. We'll begin with another example of a simple hormone pathway, the regulation of calcium ion concentration in the circulatory system.

Parathyroid Hormone and Vitamin D: Control of Blood Calcium

Because calcium ions (Ca^{2+}) are essential to the normal functioning of all cells, homeostatic control of the level of calcium in the blood is vital. If the blood Ca^{2+} level falls substantially, skeletal muscles begin to contract convulsively, a potentially fatal

condition. If the blood Ca^{2+} level rises substantially, calcium phosphate can form precipitates in body tissues, leading to widespread organ damage.

In mammals, the **parathyroid glands**, a set of four small structures embedded in the posterior surface of the thyroid (see Figure 45.8), play a major role in blood Ca^{2+} regulation. When the blood Ca^{2+} level falls below a set point of about 10 mg/100 mL, these glands release **parathyroid hormone (PTH)**.

PTH raises the level of blood Ca^{2+} through direct effects in bones and the kidneys and an indirect effect on the intestines (**Figure 45.18**). In bones, PTH causes the mineralized matrix to break down, releasing Ca^{2+} into the blood. In the kidneys, PTH directly stimulates reabsorption of Ca^{2+} through the renal tubules. In addition, PTH indirectly raises the blood Ca^{2+} level by promoting production of vitamin D. A precursor form of vitamin D is obtained from food or synthesized by skin exposed to sunlight. Conversion of this precursor to active vitamin D begins in the liver. PTH acts in the kidney to stimulate completion of the conversion process. Vitamin D in turn acts on the intestines, stimulating the uptake of Ca^{2+} from food. As the blood Ca^{2+} level rises, a negative-feedback loop inhibits further release of PTH from the parathyroid glands (not shown in Figure 45.18).

The thyroid gland can also contribute to calcium homeostasis. If the blood Ca^{2+} level rises above the set point, the thyroid gland releases **calcitonin**, a hormone that inhibits bone breakdown and enhances Ca^{2+} excretion by the kidneys. In fishes, rodents, and some other animals, calcitonin is required for Ca^{2+} homeostasis. In humans, however, calcitonin is apparently needed only during the extensive bone growth of childhood.

► **Figure 45.18** The roles of parathyroid hormone (PTH) in regulating the blood calcium level in mammals.

